

Polygon Lattice Points

Time limit: 1.00 s

Memory limit: 512 MB

Given a polygon, your task is to calculate the number of lattice points inside the polygon and on its boundary. A lattice point is a point whose coordinates are integers.

The polygon consists of n vertices $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The vertices (x_i, y_i) and (x_{i+1}, y_{i+1}) are adjacent for $i = 1, 2, \dots, n-1$, and the vertices (x_1, y_1) and (x_n, y_n) are also adjacent.

Input

The first input line has an integer n : the number of vertices.

After this, there are n lines that describe the vertices. The i th such line has two integers x_i and y_i .

You may assume that the polygon is simple, i.e., it does not intersect itself.

Output

Print two integers: the number of lattice points inside the polygon and on its boundary.

Constraints

- $3 \leq n \leq 10^5$
- $-10^9 \leq x_i, y_i \leq 10^9$

Example

Input	Output
4 1 1 5 3 3 5 1 4	6 8